

Akshata Madavi

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Summary

MS Software Engineering student at SJSU with 6 years of production data engineering and hands-on AI/ML shipped in grad school and industry. As an Applied AI Engineer Intern at The Mortgage Office, built a document understanding microservice (OCR, layout, extraction) that cut recurring subscription costs and gave the team full pipeline control, plus evaluation graders that catch agent failures before they ship and cut the release review cycle from weeks to days. Also trained a real-time anomaly detection model (XGBoost, 98% accuracy) and fine-tuned LLMs via LoRA (+10 pp accuracy). Brings the rare combination of large-scale data infrastructure depth and modern AI/ML engineering.

Education

San Jose State University, MS in Software Engineering May 2027

Machine Learning, Deep Learning, Recommender Systems, Software Systems Engineering

College of Engineering Pune (COEP), BS in Computer Engineering May 2018

Skills

- **Languages:** Python, SQL
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, PySpark ML, CNNs; Evaluation
- **LLMs & GenAI:** Fine-tuning/LoRA; RAG(LangChain, LlamaIndex); Vector DB; LangSmith; LLM Serving(Vertex AI)
- **Data & Streaming:** Kafka/KSQL, Spark, BigQuery, Airflow, Pandas, NumPy, MySQL
- **MLOps & Cloud:** MLflow, Docker, GitHub Actions CI/CD, GCP (Vertex AI, BigQuery), AWS, Azure

Work Experience

Applied AI Engineer Intern, The Mortgage Office – San Mateo, CA Jul 2026 – Present

- Built and productionized a **document understanding microservice** handling OCR, layout parsing, and information extraction end-to-end on arbitrary documents; replacing a paid subscription **cut recurring subscription costs**, put the pipeline fully under the team's control, and improved accuracy across all downstream consumers.
- Built **code-based** and **LLM-as-judge evaluation graders** that systematically surface production LLM agent failures (e.g., incorrect outputs, tool calls missed on **~50% of turns**) before they reach users — cutting the release review cycle from **weeks to days** and giving the team a repeatable way to ship agent changes with confidence.

Senior Data Engineer, LTIMindtree – Pune, India May 2023 – Mar 2025

- Engineered near real-time GCP/BigQuery data pipelines ingesting **~10M records** from enterprise sources, building ML-ready feature tables for downstream AI/ML models at production scale.
- Redesigned a multi-source ETL service (~30 GB) using parallel Spark jobs, cutting ingestion from **1 week to under 10 minutes**; served as the primary feature store for ML workloads.

Data Engineer, AI Adventures – Pune, India Aug 2021 – Apr 2023

- Led migration of course data from Moodle to MySQL; A/B testing and retention analysis drove a **20% revenue increase**.

Data Engineer, STL (Sterlite Technologies) – Pune, India Jun 2018 – May 2021

- Built near real-time GCP pipelines migrating Oracle, SAP, and Salesforce to BigQuery (**~1 TB**); applied DOE-based statistical analysis for manufacturing scrap detection and process optimization.
- Developed automated anomaly detection checks over sensor and telemetry outputs, reducing manual review time and catching production anomalies earlier.

Projects

VitalWatch – Real-Time Sensor Anomaly Detection (AI-Catalyst Hackathon 2025)

PyTorch, XGBoost, Kafka/KSQL, GCP Vertex AI, BigQuery, Twilio, Python

- Trained XGBoost on multi-channel time-series sensor data achieving **98% anomaly detection accuracy**; deployed on GCP Vertex AI.
- Built Kafka/KSQL streaming pipeline enriching flagged events via LLM with BigQuery data; delivered end-to-end alerts in **~10 seconds**.

Personal Finance Auditor – Agentic Text-to-SQL Assistant

PydanticAI, Gemini, SQLite, sqlglot, Streamlit, Logfire, Python

- Built a multi-agent pipeline: parallel LLM agents categorize bank statement merchants at ingest; a conversational agent answers plain-English spending questions via a Text-to-SQL tool loop against SQLite.
- Enforced two-layer SQL safety guardrail (sqlglot parse-level SELECT validation + read-only SQLite connection); instrumented all agent runs with Logfire traces (LLM calls, tool calls, latency, tokens); deployed via Streamlit UI.

Enterprise AI Governance Platform – LLM Fine-tuning & Monitoring

PyTorch, HuggingFace, LoRA, LangChain, FAISS, MLflow, AWS, Docker

- Fine-tuned Qwen 0.6B via LoRA; tracked experiments with MLflow improving accuracy from **87% to 97% (+10 pp)**.
- Designed LangChain + FAISS retrieval pipelines for failure pattern analysis; containerized with Docker for reproducible deployment.